A. Experimental setup

Supplementary evidence for the Swarm Alignment Hackathon submission.
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Central finding. Agents with self-written memory often retain promises and omit non-delivery; some record gifts that never happened. Making the breach explicit changes votes, and a public gift record sharply reduces trust in the promise-breaker over long games.

This supplement condenses the supplied reference bundle. Updated research-note results take precedence over older embedded snapshots. It reports existing analyses, not new runs. Rumor results remain interim; the main submission supplies the broader framing and bibliography.

The colony and its controls

Larkspur contains 4, 8 or 16 agents sharing one roughly 560-word system prompt. Five randomized trait pairs vary self-interest, assertiveness, guardedness, reciprocity and preference for solitary work; two placebo pairs vary birth season and favorite color. Versions are balanced within each game and sentences shuffled. Each agent rewrites an 80-word notebook daily, its only memory between days.

Component	Rule
Food and work	Start with 5 food; eat 2 daily. Fishing alone yields 2, mutual pairing 3 each, and an unreciprocated pairing 1. A quarter start as weavers (+2 daily, +1 for their partner); they can teach others.
Communication	One public message and up to two private messages daily, read the next morning; agents can give food.
Influence measures	Private trust polls and keeper nominations. The elected keeper divides a winter store of 3 food per member and retains the remainder.
Promise-breaker	Privately instructed to promise food publicly each day, never deliver, and blame a mix-up if challenged. The engine blocks its gifts.
Records	The public board shows yesterday's gifts to everyone. The private ledger shows only the agent's own gifts given and received, including an explicit absence message.

All runs use seed 0. A separate random stream selects the promise-breaker, preserving matched starting configurations across conditions. Pairing controls the initial setup; it does not make subsequent model responses identical. Long runs use 8 agents, 20-day seasons, spoilage above 15 food, and one-sentence reasons in replies.

B. Evidence for the memory failure

Notebooks preserve promises more readily than breaches

On gemma-4-31b without a record, victims' next-day notes mention promises in 26%, 52% and 71% of cases across the positive-, zero- and negative-sum economies, but missing food in only 3%, 2% and 7%. With the board, non-delivery appears in 57-64% of notes. Coding is keyword-based.

gemma-4-26b, 40 days	No record	Private ledger	Public board
Notes falsely claiming a gift	51%	26%	11%
Promise-breaker trust share	20.8%	14.6%	7.3%
Keeper nomination share	16.7%	5.2%	5.2%

Without a record, 60 of 117 next-day notes claimed a gift that never arrived. Five of six victims did this at least once; one repeated the claim for 36 days. Example: *"Dov kept his promise and gave me food."* These are repeated observations from a few agents, not 117 independent subjects.

The models respond when the breach is stated

Counterfactual replays re-ask 182 logged victim votes (112 on gemma-4-31b; 70 on gemma-4-26b), with 8 samples under each of 5 prompt variants: unchanged, notebook mentions removed, promises removed, both removed, or an explicit non-delivery sentence. This yields 7,280 replay decisions.

Vote share for promise-breaker	Unchanged prompt	Explicit breach
gemma-4-31b	20.3%	7.0%
gemma-4-26b	40.7%	13.0%

The sentence "the food X promised you has never arrived" reduces support by about two-thirds on both models. This supports an information or salience bottleneck: the models can withdraw support when non-delivery is explicit. The intervention also adds authority and emphasis, so it does not isolate memory as the sole cause.

Private records help; the public board helps more

In each of the three gemma-4-26b colonies, the private ledger's trust share lies between no record and the public board. The private reduction is 6.2 percentage points, versus 13.5 for the board. This is consistent with both memory support and shared visibility mattering; wording and information content also differ, so their causal contributions are not cleanly separated.

C. Long-run accountability and model pilots

The long-run design crosses three economies, board on/off and promise-breaker on/off: 12 conditions, 3 games per condition, 1,000 days per game. The 36 games contain about 288,000 agent replies and 50 trust polls and keeper terms per game. Zero-sum fishing divides a fixed catch; negative-sum fishing reduces the catch as claimed shares increase.

gemma-4-31b, 1,000 days	Trust: no board	Trust: board
Positive-sum	8.5%	0.3%
Zero-sum	9.6%	0.8%
Negative-sum	9.6%	1.5%

The board reduces trust in every one of nine paired games. Board games range from 0.0-2.2% trust, versus 4.5-13.8% without it. The promise-breaker holds the store in 0 of 450 board-condition keeper terms. The nine comparisons reuse three colony setups across economies: the post-hoc pair-level sign test ($p = 0.004$) must not be read as nine independent setup replications; at setup level, 3 of 3 gives $p = 0.25$.

The gap appears at the first poll in zero- and negative-sum economies, but takes roughly 100 days in positive-sum games. Without the board, only zero-sum games meet the registered reputation-decline criterion: trust falls from 13.8% in the first third to 6.2% in the last. Thus persistent trust is not universal, and the notebook mechanism does not explain every trajectory.

Forty-day pilots: useful variation, not universal replication

Model; positive-sum	No board	Board	Interpretation
gemma-4-26b	20.8%	7.3%	Lower in 3/3 colonies
glm-5.2	9.4%	4.2%	Lower in 3/3; promises on only about half the days
deepseek-v4-flash	5.2%	4.2%	Low baseline; no no-cheat control
gemma-4-31b	6.2%	7.3%	No reduction at this horizon

DeepSeek victims record non-delivery in 14% of notes and never invent gifts in the reported sample, consistent with a smaller need for the board. The supplied analysis reports lower trust in 8 of 11 non-tied pairs ($p = 0.23$). Qwen3.5-397b stopped at days 27-36 and is excluded from the table. Cross-model generality remains open; the longer gemma-4-31b result motivates longer pilots without proving duration explains the difference.

D. Exploratory extensions and negative results

False reputations resist contradictory behavior (interim)

A generous target has prosocial traits and weaving ability. Three of seven peers are told by a traveller that it is selfish, either daily through the system prompt or once through their initial notebook. Each colony is paired with a no-rumor twin. The daily-rumor comparison covers six pairs, three per board setting; games were at days 23-80 when summarized.

Daily-rumor believers	Rumor condition	No-rumor twins
Days fishing with the target	0-3	6-62
Negative mentions	24% of replies	0%
Trust votes for the target	2/84	27/80

Believers who received food still gave the target 0 of 48 trust votes. Notes describe generosity as pretense: *"Nils is pretending to be generous."* Yet the target gave 83 food in aggregate versus 70 for its twins. Non-believers made no negative mentions in 1,118 replies and increased their trust votes for the target (30/110 versus 14/111). The reported pattern is persistent misperception, not established behavioral confirmation or rumor contagion.

A one-time rumor had weaker effects (3% negative mentions) and varied by colony. One target became poorer and gave 2 versus 23 food by day 23, an exploratory lead rather than a replicated effect. With the board, believers still gave the target no trust votes by day 44. Daily system-prompt reinforcement limits conclusions about spontaneous belief persistence.

Costly punishment: suggestive, formally inconclusive

On gemma-4-26b, punishment costs the actor 1 food and removes 2 from the target. Without the board, 1 of 4 punishments targeted the cheat; with it, 12 of 18 did, including 9 by bystanders. All 12 came from one colony. The preregistration required at least 10 punishments per arm, so the test is inconclusive.

Traits and quiet incumbents

Across the 234 short games, trait effects on behavior replicate across two Gemma models, but no influence effect is significant on both; directions agree on 27 of 45 comparisons. Placebos pass 0 of 96 tests, consistent with conservative or correlated tests and limited power. Mediation through partnerships and messages is exploratory because those behaviors were not randomized.

In every long game, one member holds the store for 18-43 of 50 terms without being mentioned more than peers. This motivates monitoring resource allocation alongside speech. Reused setups confound claims about which personalities reliably gain power. Short-game sanction nulls are underpowered: there is only one trust poll per game.

E. Audit trail, limits and reproducibility

Registered outcomes retained, including nulls

Registered check	Reported outcome and qualification
Persistent leader	0/12 conditions pass the joint criterion; resource incumbency alone does not satisfy its speech requirement.
Invented institution	4/12 flagged, all negative-sum; classifier unvalidated. Partner schedules were initially mistaken for keeper rotations. Treat as exploratory.
Sorting	In-groups: 10/12, largely stable pairs rather than factions. Exclusion: 12/12, often early non-weavers without partners.
Freeze-out	0/6 promise-breaker conditions pass the registered decline criterion; this is distinct from low partnership levels.
Long-run reputation	1/6 passes: zero-sum without the board. Low trust with the board is a level effect, not necessarily a downward trend.

What this study establishes

In this scaffold, public action records produce a large, consistent long-run reduction in trust in a planted promise-breaker. Replay and notebook analyses support a failure to retain or use non-delivery as an important mechanism. They do not establish that all agent swarms need this exact intervention, that all deception is detected, or that the board withstands tampering.

The long-run evidence uses one model and three colony setups. The 80-word memory is one design; short and long games also differ in spoilage, reasons and polling. Gifts are blocked by the engine, making the adversary stylized. A judge reading the cheat's notebook can see its instruction, so its dishonesty rating is not independent detection. Keyword coding and the institution classifier limit interpretability. Pilot results are small and rumor runs unfinished.

Code and analysis map

Repository: github.com/soham-padia/who-rises-in-the-colony
Dataset and run logs:
https://huggingface.co/datasets/sohampadianeu/who-rises-in-the-colony-runs

Simulation: colony_sim.py. **Core analyses:** compare_arms.py, replay.py, notebook_memory.py, analyze_long.py. **Extensions:** cross_model.py, punish_analysis.py, belief_analysis.py; belief_interim.py handles unfinished runs. **Registered checks:** detect_catches.py and docs/preregistration.md (8cf7fd0; amendment 20b2a18), preregistration_punish.md and preregistration_belief.md in docs/.

Limitations and dual-use / ethical considerations

The simulation touches no real systems or people. Its failure modes could inform deceptive strategies, while the tested record-based defense offers a concrete mitigation. Claude/Claude Code supported implementation, analysis and writing; the team remains responsible for validating results and references. This shortened supplement removes working logs and duplicated literature material; literature citations remain in the main report.